

Risk and Review Evaluation Design

Pre-registered thresholds and metrics for calibration, safety, and review efficiency.

Applies to all six task families

	Control Knob (Pre-registered)	Purpose	Primary Metrics	Definition (Operational)
1	 Confidence Floor = 0.60	→ Prevent overconfident outputs and reduce unsafe autonomy.	- ECE (Expected Calibration Error) ↓ (target: 0.10) - False Autonomy Rate ↓ (fraction of overconfident & incorrect)	All model output confidence used for actions or answers must be 0.60. Below this, the system must defer, request more evidence, or escalate.
2	 Reinspection Trigger = 0.40	→ Ensure reinspection when uncertainty is high or coverage is insufficient.	- Mission Validity Rate ↑ (successful reinspection missions / total) - ROI Coverage Gain ↑ (A ROI coverage after reinspection)	Trigger reinspection when predicted confidence < 0.40 or ROI coverages below requirement for the task.
3	 Review Budget Cap = 0.25 (25% max)	→ Limit unnecessary human review and control operational cost.	- Review Rate ↓ (human-reviewed items / total) - Unnecessary Review Rate ↓ (human agrees & confidence ≥ 0.60)	At most 25% of items may be escalated to human review. Exceeding requires justification. Prefer abstain or gather more evidence instead.
4	 Tool / Runtime Budget = 12 calls / 300s	→ Bound agent cost, latency, and infrastructure exposure.	- Latency (p50/ p95) ↓ (seconds per task) - Infra Failure Rate ↓ (timeout / error / tool failure)	At most 25% of items may be escalated to human review. Exceeding requires justification. Prefer abstain or gather more evidence instead.

Global Evaluation Summary (reported per task family and overall)

 ECE ↓
Calibration

 False Autonomy ↓
Safety

 Mission Validity ↓
Reinspection Effectiveness

 Review Rate ↓
Efficiency

 Unnecessary Review ↓
Human Load

 Latency ↓
Speed

 Infra Failure ↓
Reliability